

May 18, 2016

Ms. Amy Vail  
City of Cadillac WWTP  
200 N. Lake St.  
Cadillac, MI 49601

Subject: Executive Summary for Project Number 208079

Dear Ms. Vail:

Please find the attached biomonitoring report for testing conducted May 05, 2016 through May 12, 2016. The results for this evaluation are:

| Test Species       | TUa | TUc | MATC |
|--------------------|-----|-----|------|
| Ceriodaphnia dubia | 0   | 0   | >100 |
| Fathead Minnows    | 0   | 0   | >100 |

The exposure tests were valid since they met the criteria for survival, reproduction, and growth in the controls. Should you have any questions pertaining to this report please let me know.

Sincerely,  
Michael Amboyan  
Paragon Laboratories, Inc.

**DEQ**  
**MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY - WATER BUREAU**  
**CERIODAPHNIA DUBIA CHRONIC TOXICITY TEST REPORT**  
*By authority of PA 451 of 1994, as amended.*

|                                                                                                                                                                   |       |                                                    |                   |                                                                                                                    |                                                                   |                                                                              |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------|--|
| 1. NAME OF FACILITY (on NPDES permit)<br>Cadillac WWTP                                                                                                            |       |                                                    |                   | 2. NPDES PERMIT #<br>MI0020257                                                                                     |                                                                   |                                                                              |  |
| 3. RECEIVING WATER (as designated in permit)<br>Saline River                                                                                                      |       |                                                    | 4. OUTFALL<br>001 |                                                                                                                    | 5. RECEIVING WATER CONCENTRATION (if known)<br>Unknown-Cerio + FH |                                                                              |  |
| 6. TEST LAB (Name and Address)<br>Paragon Laboratories, Inc. 12649 Richfield Ct. Livonia MI 48150                                                                 |       |                                                    |                   |                                                                                                                    |                                                                   |                                                                              |  |
| 7. TEST START DATE<br>05/05/16                                                                                                                                    |       | 8. TEST END DATE<br>05/12/16                       |                   | 9. AGE RANGE OF ORGANISMS AT TEST START<br>Less Than 24 Hours                                                      |                                                                   | 10. Report Date<br>5/18/2016 9:02:24 AM                                      |  |
| 11. NAME OF PERSON CONDUCTING TEST<br>Michael Amboyan                                                                                                             |       |                                                    |                   | 12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT<br>Paragon Laboratories, Inc. (734) 462-3900 |                                                                   |                                                                              |  |
| 13. SAMPLE COLLECTION DATES<br>5/4/2016 7:00:00 AM                                                                                                                |       | 14. DATE RECEIVED<br>5/5/2016 9:32:00 AM           |                   | 15. DATE OF FIRST USE<br>05/05/2016                                                                                |                                                                   | 16. ARRIVAL TEMP (°C)<br>Sample 1: 4 on ice                                  |  |
| 5/5/2016 7:00:00 AM                                                                                                                                               |       | 5/6/2016 9:30:00 AM                                |                   | 05/07/2016                                                                                                         |                                                                   | Sample 2: 4 on ice                                                           |  |
| 5/9/2016 7:00:00 AM                                                                                                                                               |       | 5/10/2016 9:50:00 AM                               |                   | 05/10/2016                                                                                                         |                                                                   | Sample 3: 3 on ice                                                           |  |
| 17. TOTAL RESIDUAL CHLORINE<br>Sample 1: <0.02                                                                                                                    |       | 18. AMMONIA (mg/L as N)<br>Sample 1: <0.5          |                   | 19. pH<br>Sample 1: 7.38                                                                                           |                                                                   | 20. Alkalinity<br>Sample 1: 146.5                                            |  |
| Sample 2: <0.02                                                                                                                                                   |       | Sample 2: <0.5                                     |                   | Sample 2: 7.90                                                                                                     |                                                                   | Sample 2: 161.0                                                              |  |
| Sample 3: <0.02                                                                                                                                                   |       | Sample 3: <0.5                                     |                   | Sample 3: 7.45                                                                                                     |                                                                   | Sample 3: 99.4                                                               |  |
| 21. Hardness<br>Sample 1: 181.6                                                                                                                                   |       |                                                    |                   |                                                                                                                    |                                                                   |                                                                              |  |
| Sample 2: 179.8                                                                                                                                                   |       |                                                    |                   |                                                                                                                    |                                                                   |                                                                              |  |
| Sample 3: 186.8                                                                                                                                                   |       |                                                    |                   |                                                                                                                    |                                                                   |                                                                              |  |
| 22. WAS SAMPLE DECHLORINATED?<br>Sample 1: No<br>Sample 2: No<br>Sample 3: No                                                                                     |       | 23. DESCRIBE DECHLORINATION (if any)<br>N/A        |                   |                                                                                                                    |                                                                   |                                                                              |  |
| 24. EFFLUENT SAMPLES WERE COLLECTED:<br>Facility Does Not Chlorinate                                                                                              |       |                                                    |                   |                                                                                                                    |                                                                   |                                                                              |  |
| 25. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded hold time.)<br>N/A |       |                                                    |                   |                                                                                                                    |                                                                   |                                                                              |  |
| 26. EFFLUENT FILTERED?<br>No                                                                                                                                      |       | 27. STATE MESH SIZE OF FILTER (if filtered)<br>N/A |                   |                                                                                                                    |                                                                   |                                                                              |  |
| 28. EFFLUENT SAMPLE TYPE:<br>Sample 1: 24-Hour Composite<br>Sample 2: 24-Hour Composite<br>Sample 3: 24-Hour Composite                                            |       |                                                    |                   |                                                                                                                    |                                                                   | 29. IDENTIFY THE DILUENT (O1 )<br>MHW<br>IDENTIFY THE SECONDARY (O2 )<br>N/A |  |
| 30. SUMMARY OF DATA AND RESULTS - SURVIVAL AND REPRODUCITON                                                                                                       |       |                                                    |                   |                                                                                                                    |                                                                   |                                                                              |  |
| CONCENTRATION OF EFFLUENT (%)                                                                                                                                     | O1    | 5                                                  | 15                | 30                                                                                                                 | 60                                                                | 100                                                                          |  |
| 48-HOUR SURVIVAL (%)                                                                                                                                              | 100   | 100                                                | 100               | 100                                                                                                                | 100                                                               | 100                                                                          |  |
| 7-Day MEAN REPRODUCTION/FEMALE                                                                                                                                    | 35.4  | 35.1                                               | 34.8              | 33.1                                                                                                               | 36.7                                                              | 32.9                                                                         |  |
| 7-Day MEAN SURVIVAL (%)                                                                                                                                           | 100.0 | 100.0                                              | 100.0             | 100.0                                                                                                              | 100.0                                                             | 100.0                                                                        |  |
| 31. 48-HOUR LC50 (%)<br>>100                                                                                                                                      |       | 32. TUa (acute toxic units)<br>0                   |                   |                                                                                                                    |                                                                   |                                                                              |  |
| 33. 7-Day CHRONIC VALUE (%)<br>>100                                                                                                                               |       | 34. NOEC<br>100                                    |                   | 35. LOEC<br>>100                                                                                                   |                                                                   | 36. TUc (chronic toxic units)<br>0                                           |  |

Prepared by Paragon Laboratories, Inc. Livonia MI

### FATHEAD MINNOW CHRONIC TOXICITY TEST REPORT

*By authority of PA 451 of 1994, as amended.*

|                                                                                                                                                                       |        |                                                    |                   |                                                                                                                    |                                                                   |                                                                              |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------|--|
| 1. NAME OF FACILITY (on NPDES permit)<br>Cadillac WWTP                                                                                                                |        |                                                    |                   | 2. NPDES PERMIT #<br>MI0020257                                                                                     |                                                                   |                                                                              |  |
| 3. RECEIVING WATER (as designated in permit)<br>Saline River                                                                                                          |        |                                                    | 4. OUTFALL<br>001 |                                                                                                                    | 5. RECEIVING WATER CONCENTRATION (if known)<br>Unknown-Cerio + FH |                                                                              |  |
| 6. TEST LAB (Name and Address)<br>Paragon Laboratories, Inc. 12649 Richfield Ct. Livonia MI 48150                                                                     |        |                                                    |                   |                                                                                                                    |                                                                   |                                                                              |  |
| 7. TEST START DATE<br>05/05/16                                                                                                                                        |        | 8. TEST END DATE<br>05/12/16                       |                   | 9. AGE RANGE OF ORGANISMS AT TEST START<br>Less Than 24 Hours                                                      |                                                                   | 10. Report Date<br>5/18/2016 9:02:24 AM                                      |  |
| 11. NAME OF PERSON CONDUCTING TEST<br>Michael Amboyan                                                                                                                 |        |                                                    |                   | 12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT<br>Paragon Laboratories, Inc. (734) 462-3900 |                                                                   |                                                                              |  |
| 13. SAMPLE COLLECTION DATES<br>5/4/2016 7:00:00 AM                                                                                                                    |        | 14. DATE RECEIVED<br>5/5/2016 9:32:00 AM           |                   | 15. DATE OF FIRST USE<br>05/05/2016                                                                                |                                                                   | 16. ARRIVAL TEMP (°C)<br>Sample 1: 4 on ice                                  |  |
| 5/5/2016 7:00:00 AM                                                                                                                                                   |        | 5/6/2016 9:30:00 AM                                |                   | 05/07/2016                                                                                                         |                                                                   | Sample 2: 4 on ice                                                           |  |
| 5/9/2016 7:00:00 AM                                                                                                                                                   |        | 5/10/2016 9:50:00 AM                               |                   | 05/10/2016                                                                                                         |                                                                   | Sample 3: 3 on ice                                                           |  |
| 17. TOTAL RESIDUAL CHLORINE<br>Sample 1: <0.02                                                                                                                        |        | 18. AMMONIA (mg/L as N)<br>Sample 1: <0.5          |                   | 19. pH<br>Sample 1: 7.38                                                                                           |                                                                   | 20. Alkalinity<br>Sample 1: 146.5                                            |  |
| Sample 2: <0.02                                                                                                                                                       |        | Sample 2: <0.5                                     |                   | Sample 2: 7.90                                                                                                     |                                                                   | Sample 2: 161.0                                                              |  |
| Sample 3: <0.02                                                                                                                                                       |        | Sample 3: <0.5                                     |                   | Sample 3: 7.45                                                                                                     |                                                                   | Sample 3: 99.4                                                               |  |
| 21. Hardness<br>Sample 1: 181.6                                                                                                                                       |        | Sample 2: 179.8                                    |                   | Sample 3: 186.8                                                                                                    |                                                                   |                                                                              |  |
| 22. WAS SAMPLE DECHLORINATED?<br>Sample 1: No<br>Sample 2: No<br>Sample 3: No                                                                                         |        | 23. DESCRIBE DECHLORINATION (if any)<br><br>N/A    |                   |                                                                                                                    |                                                                   |                                                                              |  |
| 24. EFFLUENT SAMPLES WERE COLLECTED:<br>Facility Does Not Chlorinate                                                                                                  |        |                                                    |                   |                                                                                                                    |                                                                   |                                                                              |  |
| 25. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded hold time.)<br><br>N/A |        |                                                    |                   |                                                                                                                    |                                                                   |                                                                              |  |
| 26. EFFLUENT FILTERED?<br>No                                                                                                                                          |        | 27. STATE MESH SIZE OF FILTER (if filtered)<br>N/A |                   |                                                                                                                    |                                                                   |                                                                              |  |
| 28. EFFLUENT SAMPLE TYPE:<br>Sample 1: 24-Hour Composite<br>Sample 2: 24-Hour Composite<br>Sample 3: 24-Hour Composite                                                |        |                                                    |                   |                                                                                                                    |                                                                   | 29. IDENTIFY THE DILUENT (O1 )<br>MHW<br>IDENTIFY THE SECONDARY (O2 )<br>N/A |  |
| <b>30. SUMMARY OF DATA AND RESULTS - SURVIVAL AND REPRODUCITON</b>                                                                                                    |        |                                                    |                   |                                                                                                                    |                                                                   |                                                                              |  |
| CONCENTRATION OF EFFLUENT (%)                                                                                                                                         | O1     | 5                                                  | 15                | 30                                                                                                                 | 60                                                                | 100                                                                          |  |
| 96-HOUR SURVIVAL (%)                                                                                                                                                  | 97.5   | 97.5                                               | 92.5              | 97.5                                                                                                               | 95                                                                | 97.5                                                                         |  |
| 7-DAY MEAN BIOMASS (mg/initial fish)                                                                                                                                  | 0.3664 | 0.4328                                             | 0.1447            | 0.4290                                                                                                             | 0.4018                                                            | 0.4123                                                                       |  |
| 7-DAY MEAN SURVIVAL (%)                                                                                                                                               | 97.5   | 97.5                                               | 92.5              | 97.5                                                                                                               | 92.5                                                              | 95                                                                           |  |
| 31. 96-HOUR LC50 (%)<br>>100                                                                                                                                          |        | 32. TUa (acute toxic units)<br>0                   |                   |                                                                                                                    |                                                                   |                                                                              |  |
| 33. 7-DAY CHRONIC VALUE (%)<br>>100                                                                                                                                   |        | 34. NOEC<br>100                                    |                   | 35. LOEC<br>>100                                                                                                   |                                                                   | 36. TUc (chronic toxic units)<br>0                                           |  |

**TOXICITY EVALUATION**  
**Ceriodaphnia Dubia Summary Table**

Client: City of Cadillac WWTP  
 Project Number: 208079  
 Primary Analyst: Michael Amboyan

| Effluent<br>Concentration | TOTAL LIVE YOUNG IN FIRST 3 BROODS PER REPLICATE |    |    |    |    |    |    |    |    |    | Total<br>No. of<br>Young | Total<br>No. of<br>Adults |
|---------------------------|--------------------------------------------------|----|----|----|----|----|----|----|----|----|--------------------------|---------------------------|
|                           | Replicate Number                                 |    |    |    |    |    |    |    |    |    |                          |                           |
|                           | 1                                                | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |                          |                           |
| 0%                        | 31                                               | 38 | 28 | 38 | 33 | 37 | 37 | 35 | 40 | 37 | 354                      | 10                        |
|                           |                                                  |    |    |    |    |    |    |    |    |    |                          |                           |
| 5%                        | 38                                               | 32 | 39 | 42 | 34 | 34 | 32 | 34 | 37 | 29 | 351                      | 10                        |
|                           |                                                  |    |    |    |    |    |    |    |    |    |                          |                           |
| 15%                       | 31                                               | 37 | 35 | 33 | 35 | 35 | 34 | 33 | 34 | 41 | 348                      | 10                        |
|                           |                                                  |    |    |    |    |    |    |    |    |    |                          |                           |
| 30%                       | 32                                               | 31 | 31 | 30 | 44 | 39 | 32 | 28 | 37 | 27 | 331                      | 10                        |
|                           |                                                  |    |    |    |    |    |    |    |    |    |                          |                           |
| 60%                       | 38                                               | 38 | 42 | 39 | 33 | 35 | 37 | 37 | 37 | 31 | 367                      | 10                        |
|                           |                                                  |    |    |    |    |    |    |    |    |    |                          |                           |
| 100%                      | 32                                               | 32 | 31 | 37 | 27 | 35 | 31 | 37 | 33 | 34 | 329                      | 10                        |
|                           |                                                  |    |    |    |    |    |    |    |    |    |                          |                           |

Qualifier Key:

A = Natural Death  
 B = Abnormal appearance of organism.  
 C = Male.  
 D = No developing young visible in brood pouch throughout test.  
 E = Aborted young.  
 F = Young produced but dead at end of the 24 hour period.  
 G = Young produced in the brood pouch but not released  
 H = Transfer death.

**TOXICITY EVALUATION  
Fathead Minnow Summary Table**

Client: City of Cadillac WWTP  
Project Number: 208079  
Primary Analyst: Michael Amboyan

| Conc. | Rep. No. | Day 7 Survivability | Initial Wt. (g) | Final Wt. (g) | Net Wt. (mg) | Avg. Wt. (mg) |
|-------|----------|---------------------|-----------------|---------------|--------------|---------------|
| 0%    | 1        | 10                  | 0.4186          | 0.42192       | 3.32         | 0.33          |
|       |          |                     |                 |               |              |               |
|       | 2        | 10                  | 0.41494         | 0.41923       | 4.29         | 0.43          |
|       |          |                     |                 |               |              |               |
|       | 3        | 10                  | 0.42108         | 0.42459       | 3.51         | 0.35          |
|       |          |                     |                 |               |              |               |
| 5%    | 5        | 10                  | 0.41224         | 0.41645       | 4.21         | 0.42          |
|       |          |                     |                 |               |              |               |
|       | 6        | 10                  | 0.41531         | 0.41976       | 4.45         | 0.44          |
|       |          |                     |                 |               |              |               |
|       | 7        | 9                   | 0.41953         | 0.42404       | 4.51         | 0.45          |
|       |          | A                   |                 |               |              |               |
| 15%   | 9        | 10                  | 0.41078         | 0.41492       | 4.14         | 0.41          |
|       |          |                     |                 |               |              |               |
|       | 10       | 8                   | 0.41626         | 0.41201       | -4.25        | -0.43         |
|       |          | A                   |                 |               |              |               |
|       | 11       | 10                  | 0.41773         | 0.4225        | 4.77         | 0.48          |
|       |          |                     |                 |               |              |               |
|       | 12       | 10                  | 0.41675         | 0.42119       | 4.44         | 0.44          |
|       |          |                     |                 |               |              |               |
|       | 12       | 9                   | 0.4157          | 0.41653       | 0.83         | 0.08          |
|       |          | A                   |                 |               |              |               |

Qualifier Key:

A = Natural Death  
B = Fungal filaments present  
C = Lethargy  
D = Excessive ventilation  
E = Erratic or uncoordinated swimming  
F = Transfer death

**TOXICITY EVALUATION  
Fathead Minnow Summary Table**

Client: City of Cadillac WWTP

Project Number: 208079

Primary Analyst: Michael Amboyan

|      |    |    |         |         |      |      |
|------|----|----|---------|---------|------|------|
| 30%  | 13 | 9  | 0.41418 | 0.41808 | 3.9  | 0.39 |
|      |    | A  |         |         |      |      |
|      | 14 | 10 | 0.41308 | 0.41789 | 4.81 | 0.48 |
|      |    |    |         |         |      |      |
|      | 15 | 10 | 0.41579 | 0.42028 | 4.49 | 0.45 |
|      |    |    |         |         |      |      |
|      | 16 | 10 | 0.42453 | 0.42849 | 3.96 | 0.4  |
|      |    |    |         |         |      |      |
| 60%  | 17 | 10 | 0.41184 | 0.41564 | 3.8  | 0.38 |
|      |    |    |         |         |      |      |
|      | 18 | 9  | 0.41364 | 0.41748 | 3.84 | 0.38 |
|      |    | A  |         |         |      |      |
|      | 19 | 10 | 0.41994 | 0.42419 | 4.25 | 0.42 |
|      |    |    |         |         |      |      |
|      | 20 | 8  | 0.41835 | 0.42253 | 4.18 | 0.42 |
|      |    | A  |         |         |      |      |
| 100% | 21 | 9  | 0.42258 | 0.42666 | 4.08 | 0.41 |
|      |    | A  |         |         |      |      |
|      | 22 | 9  | 0.41123 | 0.41524 | 4.01 | 0.4  |
|      |    | A  |         |         |      |      |
|      | 23 | 10 | 0.41725 | 0.42173 | 4.48 | 0.45 |
|      |    |    |         |         |      |      |
|      | 24 | 10 | 0.41562 | 0.41954 | 3.92 | 0.39 |
|      |    |    |         |         |      |      |

Qualifier Key:

A = Natural Death

B = Fungal filaments present

C = Lethargy

D = Excessive ventilation

E = Erratic or uncoordinated swimming

F = Transfer death

**TOXICITY EVALUATION**  
**Ceriodaphnia Survival and Reproduction Test**

Client: City of Cadillac WWTP

Project Number: 208079

Primary Analyst: Michael Amboyan

| Concentration | Day   | Replicate |    |    |    |    |    |    |    |    |    | No. of Young | No. of Adults | Young per Adult |
|---------------|-------|-----------|----|----|----|----|----|----|----|----|----|--------------|---------------|-----------------|
|               |       | 1         | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |              |               |                 |
| 0%            | 0*    | 1         | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0            | 10            | 0.0             |
|               | 1     | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 10            | 0.0             |
|               | 2     | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 10            | 0.0             |
|               | 3     | 2         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 2            | 10            | 0.2             |
|               | 4     | 0         | 8  | 5  | 8  | 6  | 7  | 9  | 5  | 8  | 7  | 63           | 10            | 6.3             |
|               | 5     | 12        | 12 | 10 | 11 | 8  | 12 | 10 | 12 | 13 | 13 | 113          | 10            | 11.3            |
|               | 6     | 17        | 18 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 17 | 52           | 10            | 5.2             |
|               | 7     | 0         | 0  | 13 | 19 | 19 | 18 | 18 | 18 | 19 | 0  | 124          | 10            | 12.4            |
|               | 8     |           |    |    |    |    |    |    |    |    |    |              |               |                 |
|               | Total | 31        | 38 | 28 | 38 | 33 | 37 | 37 | 35 | 40 | 37 | 354          | 10            | 35.4            |

| Concentration | Day   | Replicate |    |    |    |    |    |    |    |    |    | No. of Young | No. of Adults | Young per Adult |
|---------------|-------|-----------|----|----|----|----|----|----|----|----|----|--------------|---------------|-----------------|
|               |       | 11        | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |              |               |                 |
| 5%            | 0*    | 1         | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0            | 10            | 0.0             |
|               | 1     | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 10            | 0.0             |
|               | 2     | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 10            | 0.0             |
|               | 3     | 5         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 5            | 10            | 0.5             |
|               | 4     | 0         | 4  | 7  | 6  | 4  | 6  | 5  | 6  | 6  | 5  | 49           | 10            | 4.9             |
|               | 5     | 10        | 10 | 11 | 15 | 12 | 13 | 10 | 10 | 11 | 10 | 112          | 10            | 11.2            |
|               | 6     | 23        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 14 | 37           | 10            | 3.7             |
|               | 7     | 0         | 18 | 21 | 21 | 18 | 15 | 17 | 18 | 20 | 0  | 148          | 10            | 14.8            |
|               | 8     |           |    |    |    |    |    |    |    |    |    |              |               |                 |
|               | Total | 38        | 32 | 39 | 42 | 34 | 34 | 32 | 34 | 37 | 29 | 351          | 10            | 35.1            |

\*Neonates less than 24 hours old added to each replicate vessel at start of test. Values are not included in replicate total.

**TOXICITY EVALUATION**  
**Ceriodaphnia Survival and Reproduction Test**

Client: City of Cadillac WWTP

Project Number: 208079

Primary Analyst: Michael Amboyan

| Concentration | Day   | Replicate |    |    |    |    |    |    |    |    |    | No. of Young | No. of Adults | Young per Adult |
|---------------|-------|-----------|----|----|----|----|----|----|----|----|----|--------------|---------------|-----------------|
|               |       | 21        | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |              |               |                 |
| 15%           | 0*    | 1         | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0            | 10            | 0.0             |
|               | 1     | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 10            | 0.0             |
|               | 2     | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 10            | 0.0             |
|               | 3     | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 5  | 5            | 10            | 0.5             |
|               | 4     | 6         | 6  | 9  | 7  | 6  | 7  | 5  | 7  | 6  | 0  | 59           | 10            | 5.9             |
|               | 5     | 7         | 8  | 11 | 8  | 9  | 10 | 12 | 8  | 10 | 12 | 95           | 10            | 9.5             |
|               | 6     | 18        | 23 | 0  | 18 | 0  | 0  | 0  | 0  | 0  | 24 | 83           | 10            | 8.3             |
|               | 7     | 0         | 0  | 15 | 0  | 20 | 18 | 17 | 18 | 18 | 0  | 106          | 10            | 10.6            |
|               | 8     |           |    |    |    |    |    |    |    |    |    |              |               |                 |
|               | Total | 31        | 37 | 35 | 33 | 35 | 35 | 34 | 33 | 34 | 41 | 348          | 10            | 34.8            |

| Concentration | Day   | Replicate |    |    |    |    |    |    |    |    |    | No. of Young | No. of Adults | Young per Adult |
|---------------|-------|-----------|----|----|----|----|----|----|----|----|----|--------------|---------------|-----------------|
|               |       | 31        | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |              |               |                 |
| 30%           | 0*    | 1         | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0            | 10            | 0.0             |
|               | 1     | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 10            | 0.0             |
|               | 2     | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 10            | 0.0             |
|               | 3     | 0         | 0  | 2  | 0  | 0  | 0  | 4  | 0  | 0  | 0  | 6            | 10            | 0.6             |
|               | 4     | 4         | 7  | 0  | 6  | 7  | 7  | 0  | 6  | 7  | 3  | 47           | 10            | 4.7             |
|               | 5     | 9         | 10 | 12 | 10 | 15 | 13 | 12 | 12 | 12 | 12 | 117          | 10            | 11.7            |
|               | 6     | 19        | 14 | 17 | 14 | 0  | 0  | 16 | 0  | 18 | 0  | 98           | 10            | 9.8             |
|               | 7     | 0         | 0  | 0  | 0  | 22 | 19 | 0  | 10 | 0  | 12 | 63           | 10            | 6.3             |
|               | 8     |           |    |    |    |    |    |    |    |    |    |              |               |                 |
|               | Total | 32        | 31 | 31 | 30 | 44 | 39 | 32 | 28 | 37 | 27 | 331          | 10            | 33.1            |

\*Neonates less than 24 hours old added to each replicate vessel at start of test. Values are not included in replicate total.



**TOXICITY EVALUATION**  
**Ceriodaphnia Survival and Reproduction Test**

Client: City of Cadillac WWTP

Project Number: 208079

Primary Analyst: Michael Amboyan

| Concentration | Day   | Replicate |    |    |    |    |    |    |    |    |    | No. of Young | No. of Adults | Young per Adult |
|---------------|-------|-----------|----|----|----|----|----|----|----|----|----|--------------|---------------|-----------------|
|               |       | 41        | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |              |               |                 |
| 60%           | 0*    | 1         | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0            | 10            | 0.0             |
|               | 1     | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 10            | 0.0             |
|               | 2     | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 10            | 0.0             |
|               | 3     | 0         | 6  | 0  | 0  | 0  | 0  | 6  | 0  | 0  | 4  | 16           | 10            | 1.6             |
|               | 4     | 6         | 0  | 6  | 8  | 7  | 6  | 0  | 2  | 6  | 0  | 41           | 10            | 4.1             |
|               | 5     | 12        | 12 | 12 | 12 | 11 | 12 | 10 | 13 | 12 | 13 | 119          | 10            | 11.9            |
|               | 6     | 0         | 20 | 24 | 0  | 0  | 0  | 21 | 0  | 0  | 14 | 79           | 10            | 7.9             |
|               | 7     | 20        | 0  | 0  | 19 | 15 | 17 | 0  | 22 | 19 | 0  | 112          | 10            | 11.2            |
|               | 8     |           |    |    |    |    |    |    |    |    |    |              |               |                 |
|               | Total | 38        | 38 | 42 | 39 | 33 | 35 | 37 | 37 | 37 | 31 | 367          | 10            | 36.7            |

| Concentration | Day   | Replicate |    |    |    |    |    |    |    |    |    | No. of Young | No. of Adults | Young per Adult |
|---------------|-------|-----------|----|----|----|----|----|----|----|----|----|--------------|---------------|-----------------|
|               |       | 51        | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |              |               |                 |
| 100%          | 0*    | 1         | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0            | 10            | 0.0             |
|               | 1     | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 10            | 0.0             |
|               | 2     | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 10            | 0.0             |
|               | 3     | 0         | 0  | 0  | 0  | 0  | 0  | 6  | 0  | 3  | 0  | 9            | 10            | 0.9             |
|               | 4     | 6         | 6  | 7  | 6  | 6  | 6  | 0  | 8  | 0  | 5  | 50           | 10            | 5.0             |
|               | 5     | 8         | 11 | 9  | 10 | 0  | 8  | 8  | 11 | 10 | 9  | 84           | 10            | 8.4             |
|               | 6     | 18        | 15 | 0  | 0  | 0  | 0  | 17 | 0  | 20 | 0  | 70           | 10            | 7.0             |
|               | 7     | 0         | 0  | 15 | 21 | 21 | 21 | 0  | 18 | 0  | 20 | 116          | 10            | 11.6            |
|               | 8     |           |    |    |    |    |    |    |    |    |    |              |               |                 |
|               | Total | 32        | 32 | 31 | 37 | 27 | 35 | 31 | 37 | 33 | 34 | 329          | 10            | 32.9            |

\*Neonates less than 24 hours old added to each replicate vessel at start of test. Values are not included in replicate total.

**TOXICITY EVALUATION**  
**Fathead Minnow**  
**Larval Survival and Growth Test**

Client: City of Cadillac WWTP

Project Number: 208079

Primary Analyst: Michael Amboyan

| Conc. | Rep. No. | Day |    |    |    |    |    |    |    | Initial Wt. (g) | Final Wt. (g) | Net Wt. (mg) | Avg. Wt. (mg) |
|-------|----------|-----|----|----|----|----|----|----|----|-----------------|---------------|--------------|---------------|
|       |          | 0   | 1  | 2  | 3  | 4  | 5  | 6  | 7  |                 |               |              |               |
| 0%    | 1        | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.4186          | 0.42192       | 3.32         | 0.33          |
|       | 2        | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.41494         | 0.41923       | 4.29         | 0.43          |
|       | 3        | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.42108         | 0.42459       | 3.51         | 0.35          |
|       | 4        | 10  | 10 | 10 | 10 | 9  | 9  | 9  | 9  | 0.42408         | 0.42762       | 3.54         | 0.35          |

| Conc. | Rep. No. | Day |    |    |    |    |    |    |    | Initial Wt. (g) | Final Wt. (g) | Net Wt. (mg) | Avg. Wt. (mg) |
|-------|----------|-----|----|----|----|----|----|----|----|-----------------|---------------|--------------|---------------|
|       |          | 0   | 1  | 2  | 3  | 4  | 5  | 6  | 7  |                 |               |              |               |
| 5%    | 5        | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.41224         | 0.41645       | 4.21         | 0.42          |
|       | 6        | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.41531         | 0.41976       | 4.45         | 0.44          |
|       | 7        | 10  | 10 | 10 | 9  | 9  | 9  | 9  | 9  | 0.41953         | 0.42404       | 4.51         | 0.45          |
|       | 8        | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.41078         | 0.41492       | 4.14         | 0.41          |

| Conc. | Rep. No. | Day |    |    |    |    |    |    |    | Initial Wt. (g) | Final Wt. (g) | Net Wt. (mg) | Avg. Wt. (mg) |
|-------|----------|-----|----|----|----|----|----|----|----|-----------------|---------------|--------------|---------------|
|       |          | 0   | 1  | 2  | 3  | 4  | 5  | 6  | 7  |                 |               |              |               |
| 15%   | 9        | 10  | 10 | 10 | 10 | 8  | 8  | 8  | 8  | 0.41626         | 0.41201       | -4.25        | -0.43         |
|       | 10       | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.41773         | 0.4225        | 4.77         | 0.48          |
|       | 11       | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.41675         | 0.42119       | 4.44         | 0.44          |
|       | 12       | 10  | 10 | 9  | 9  | 9  | 9  | 9  | 9  | 0.4157          | 0.41653       | 0.83         | 0.08          |

**TOXICITY EVALUATION**  
**Fathead Minnow**  
**Larval Survival and Growth Test**

Client: City of Cadillac WWTP

Project Number: 208079

Primary Analyst: Michael Amboyan

| Conc. | Rep. No. | Day |    |    |    |    |    |    |    | Initial Wt. (g) | Final Wt. (g) | Net Wt. (mg) | Avg. Wt. (mg) |
|-------|----------|-----|----|----|----|----|----|----|----|-----------------|---------------|--------------|---------------|
|       |          | 0   | 1  | 2  | 3  | 4  | 5  | 6  | 7  |                 |               |              |               |
| 30%   | 13       | 10  | 10 | 10 | 10 | 9  | 9  | 9  | 9  | 0.41418         | 0.41808       | 3.9          | 0.39          |
|       | 14       | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.41308         | 0.41789       | 4.81         | 0.48          |
|       | 15       | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.41579         | 0.42028       | 4.49         | 0.45          |
|       | 16       | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.42453         | 0.42849       | 3.96         | 0.4           |

| Conc. | Rep. No. | Day |    |    |    |    |    |    |    | Initial Wt. (g) | Final Wt. (g) | Net Wt. (mg) | Avg. Wt. (mg) |
|-------|----------|-----|----|----|----|----|----|----|----|-----------------|---------------|--------------|---------------|
|       |          | 0   | 1  | 2  | 3  | 4  | 5  | 6  | 7  |                 |               |              |               |
| 60%   | 17       | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.41184         | 0.41564       | 3.8          | 0.38          |
|       | 18       | 10  | 10 | 10 | 10 | 10 | 9  | 9  | 9  | 0.41364         | 0.41748       | 3.84         | 0.38          |
|       | 19       | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.41994         | 0.42419       | 4.25         | 0.42          |
|       | 20       | 10  | 10 | 9  | 8  | 8  | 8  | 8  | 8  | 0.41835         | 0.42253       | 4.18         | 0.42          |

| Conc. | Rep. No. | Day |    |    |    |    |    |    |    | Initial Wt. (g) | Final Wt. (g) | Net Wt. (mg) | Avg. Wt. (mg) |
|-------|----------|-----|----|----|----|----|----|----|----|-----------------|---------------|--------------|---------------|
|       |          | 0   | 1  | 2  | 3  | 4  | 5  | 6  | 7  |                 |               |              |               |
| 100%  | 21       | 10  | 10 | 10 | 10 | 10 | 9  | 9  | 9  | 0.42258         | 0.42666       | 4.08         | 0.41          |
|       | 22       | 10  | 10 | 10 | 9  | 9  | 9  | 9  | 9  | 0.41123         | 0.41524       | 4.01         | 0.4           |
|       | 23       | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.41725         | 0.42173       | 4.48         | 0.45          |
|       | 24       | 10  | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 0.41562         | 0.41954       | 3.92         | 0.39          |

**TOXICITY EVALUATION**  
**Ceriodaphnia Dubia and Fathead Minnow**  
**Routine Chemical and Physical Determination**

Client: City of Cadillac WWTP

Project Number: 208079

Primary Analyst: Michael Amboyan

| Parameter | Units | Day |   |   |   |   |   |   |   |
|-----------|-------|-----|---|---|---|---|---|---|---|
|           |       | 0   | 1 | 2 | 3 | 4 | 5 | 6 | 7 |

| 0% - Initial          |          |     |     |     |     |     |     |     |  |
|-----------------------|----------|-----|-----|-----|-----|-----|-----|-----|--|
| Temperature           | °C       | 25  | 25  | 25  | 25  | 25  | 25  | 25  |  |
| Dissolved Oxygen (DO) | mg/L     | 8.6 | 8.5 | 8.0 | 9.0 | 8.8 | 8.5 | 8.4 |  |
| pH                    | S.U.     | 8.0 | 7.6 | 7.9 | 7.9 | 7.8 | 7.6 | 7.7 |  |
| Conductivity          | umhos/cm | 320 | 330 | 320 | 360 | 320 | 300 | 300 |  |

| 0% - Final           |          |     |     |     |     |     |     |     |  |
|----------------------|----------|-----|-----|-----|-----|-----|-----|-----|--|
| Temperature          | °C       | 25  | 25  | 25  | 25  | 25  | 25  | 25  |  |
| DO Cerio             | mg/L     | 8.0 | 7.8 | 9.2 | 8.6 | 8.6 | 7.5 | 7.5 |  |
| DO Fathead           | mg/L     | 8.0 | 7.0 | 7.8 | 7.8 | 6.1 | 8.0 | 0.0 |  |
| pH Cerio             | S.U.     | 8.1 | 8.0 | 8.6 | 8.6 | 8.6 | 7.6 | 7.9 |  |
| pH Fathead           | S.U.     | 7.7 | 7.9 | 7.8 | 7.8 | 7.5 | 7.8 | 0.0 |  |
| Conductivity Cerio   | umhos/cm | 320 | 320 | 480 | 530 | 490 | 340 | 360 |  |
| Conductivity Fathead | umhos/cm | 320 | 320 | 380 | 370 | 360 | 330 | 0.0 |  |

| 5% - Initial          |          |     |     |     |     |     |     |     |  |
|-----------------------|----------|-----|-----|-----|-----|-----|-----|-----|--|
| Temperature           | °C       | 25  | 25  | 25  | 25  | 25  | 25  | 25  |  |
| Dissolved Oxygen (DO) | mg/L     | 8.7 | 8.7 | 8.0 | 9.0 | 8.8 | 8.6 | 8.3 |  |
| pH                    | S.U.     | 7.7 | 7.6 | 7.7 | 7.8 | 7.9 | 7.4 | 7.7 |  |
| Conductivity          | umhos/cm | 360 | 370 | 340 | 350 | 330 | 350 | 350 |  |

| 5% - Final           |          |     |     |     |     |     |     |     |  |
|----------------------|----------|-----|-----|-----|-----|-----|-----|-----|--|
| Temperature          | °C       | 25  | 25  | 25  | 25  | 25  | 25  | 25  |  |
| DO Cerio             | mg/L     | 7.6 | 7.8 | 9.3 | 8.7 | 8.7 | 7.4 | 7.5 |  |
| DO Fathead           | mg/L     | 8.0 | 6.8 | 7.2 | 7.8 | 6.3 | 7.5 | 0.0 |  |
| pH Cerio             | S.U.     | 7.8 | 8.0 | 8.4 | 8.3 | 8.4 | 7.5 | 7.9 |  |
| pH Fathead           | S.U.     | 7.6 | 7.8 | 7.5 | 7.6 | 7.5 | 7.6 | 0.0 |  |
| Conductivity Cerio   | umhos/cm | 370 | 400 | 380 | 420 | 420 | 390 | 390 |  |
| Conductivity Fathead | umhos/cm | 370 | 380 | 360 | 380 | 360 | 370 | 0.0 |  |

**TOXICITY EVALUATION**  
**Ceriodaphnia Dubia and Fathead Minnow**  
**Routine Chemical and Physical Determination**

Client: City of Cadillac WWTP

Project Number: 208079

Primary Analyst: Michael Amboyan

| Parameter | Units | Day |   |   |   |   |   |   |   |
|-----------|-------|-----|---|---|---|---|---|---|---|
|           |       | 0   | 1 | 2 | 3 | 4 | 5 | 6 | 7 |

| 15% - Initial         |          |     |     |     |     |     |     |     |  |
|-----------------------|----------|-----|-----|-----|-----|-----|-----|-----|--|
| Temperature           | °C       | 25  | 25  | 25  | 25  | 25  | 25  | 25  |  |
| Dissolved Oxygen (DO) | mg/L     | 8.9 | 8.6 | 8.2 | 9.0 | 8.7 | 8.6 | 8.1 |  |
| pH                    | S.U.     | 7.6 | 7.7 | 7.6 | 7.8 | 7.8 | 7.4 | 7.7 |  |
| Conductivity          | umhos/cm | 440 | 450 | 450 | 460 | 420 | 450 | 460 |  |

| 15% - Final          |          |     |     |     |     |     |     |     |  |
|----------------------|----------|-----|-----|-----|-----|-----|-----|-----|--|
| Temperature          | °C       | 25  | 25  | 25  | 25  | 25  | 25  | 25  |  |
| DO Cerio             | mg/L     | 7.4 | 7.8 | 9.0 | 8.6 | 8.7 | 7.4 | 7.4 |  |
| DO Fathead           | mg/L     | 8.0 | 6.7 | 7.5 | 7.7 | 6.5 | 7.3 | 0.0 |  |
| pH Cerio             | S.U.     | 7.9 | 8.0 | 8.3 | 8.2 | 8.4 | 7.6 | 8.0 |  |
| pH Fathead           | S.U.     | 7.7 | 7.7 | 7.5 | 7.6 | 7.3 | 7.5 | 0.0 |  |
| Conductivity Cerio   | umhos/cm | 490 | 490 | 500 | 530 | 490 | 510 | 510 |  |
| Conductivity Fathead | umhos/cm | 480 | 460 | 480 | 490 | 480 | 480 | 0.0 |  |

| 30% - Initial         |          |     |     |     |     |     |     |     |  |
|-----------------------|----------|-----|-----|-----|-----|-----|-----|-----|--|
| Temperature           | °C       | 25  | 25  | 25  | 25  | 25  | 25  | 25  |  |
| Dissolved Oxygen (DO) | mg/L     | 8.7 | 8.7 | 8.0 | 9.0 | 8.7 | 8.5 | 8.2 |  |
| pH                    | S.U.     | 7.6 | 7.5 | 7.6 | 7.8 | 7.8 | 7.5 | 7.6 |  |
| Conductivity          | umhos/cm | 520 | 550 | 530 | 540 | 530 | 530 | 560 |  |

| 30% - Final          |          |     |     |     |     |     |     |     |  |
|----------------------|----------|-----|-----|-----|-----|-----|-----|-----|--|
| Temperature          | °C       | 25  | 25  | 25  | 25  | 25  | 25  | 25  |  |
| DO Cerio             | mg/L     | 7.8 | 7.8 | 9.1 | 8.5 | 8.6 | 7.5 | 7.4 |  |
| DO Fathead           | mg/L     | 8.0 | 6.9 | 7.6 | 7.8 | 6.4 | 7.6 | 0.0 |  |
| pH Cerio             | S.U.     | 7.9 | 8.0 | 8.2 | 8.1 | 8.4 | 7.6 | 8.0 |  |
| pH Fathead           | S.U.     | 7.5 | 7.6 | 7.4 | 7.5 | 7.3 | 7.5 | 0.0 |  |
| Conductivity Cerio   | umhos/cm | 540 | 590 | 590 | 620 | 610 | 590 | 620 |  |
| Conductivity Fathead | umhos/cm | 520 | 570 | 550 | 560 | 560 | 560 | 0.0 |  |

**TOXICITY EVALUATION**  
**Ceriodaphnia Dubia and Fathead Minnow**  
**Routine Chemical and Physical Determination**

Client: City of Cadillac WWTP

Project Number: 208079

Primary Analyst: Michael Amboyan

| Parameter | Units | Day |   |   |   |   |   |   |   |
|-----------|-------|-----|---|---|---|---|---|---|---|
|           |       | 0   | 1 | 2 | 3 | 4 | 5 | 6 | 7 |

| 60% - Initial         |          |     |     |     |     |     |     |     |  |
|-----------------------|----------|-----|-----|-----|-----|-----|-----|-----|--|
| Temperature           | °C       | 25  | 25  | 25  | 25  | 25  | 25  | 25  |  |
| Dissolved Oxygen (DO) | mg/L     | 8.9 | 8.6 | 8.0 | 8.9 | 8.6 | 8.7 | 8.2 |  |
| pH                    | S.U.     | 7.4 | 7.5 | 7.5 | 7.6 | 7.7 | 7.4 | 7.6 |  |
| Conductivity          | umhos/cm | 770 | 730 | 740 | 760 | 760 | 800 | 810 |  |

| 60% - Final          |          |     |     |     |     |     |     |     |  |
|----------------------|----------|-----|-----|-----|-----|-----|-----|-----|--|
| Temperature          | °C       | 25  | 25  | 25  | 25  | 25  | 25  | 25  |  |
| DO Cerio             | mg/L     | 7.9 | 7.7 | 9.1 | 8.3 | 8.6 | 7.4 | 7.5 |  |
| DO Fathead           | mg/L     | 8.0 | 7.0 | 7.7 | 7.8 | 5.8 | 7.3 | 0.0 |  |
| pH Cerio             | S.U.     | 7.8 | 7.9 | 8.2 | 8.1 | 8.3 | 7.7 | 8.1 |  |
| pH Fathead           | S.U.     | 7.6 | 7.6 | 7.4 | 7.6 | 7.3 | 7.5 | 0.0 |  |
| Conductivity Cerio   | umhos/cm | 790 | 780 | 810 | 870 | 840 | 870 | 870 |  |
| Conductivity Fathead | umhos/cm | 760 | 770 | 760 | 870 | 800 | 830 | 0.0 |  |

| 100% - Initial        |          |      |      |      |      |      |      |      |  |
|-----------------------|----------|------|------|------|------|------|------|------|--|
| Temperature           | °C       | 25   | 25   | 25   | 25   | 25   | 25   | 25   |  |
| Dissolved Oxygen (DO) | mg/L     | 8.9  | 8.7  | 8.1  | 8.9  | 8.4  | 8.6  | 8.1  |  |
| pH                    | S.U.     | 7.5  | 7.6  | 7.9  | 7.7  | 7.6  | 7.4  | 7.6  |  |
| Conductivity          | umhos/cm | 1000 | 1000 | 1100 | 1100 | 1100 | 1100 | 1100 |  |

| 100% - Final         |          |      |      |      |      |      |      |      |  |
|----------------------|----------|------|------|------|------|------|------|------|--|
| Temperature          | °C       | 25   | 25   | 25   | 25   | 25   | 25   | 25   |  |
| DO Cerio             | mg/L     | 8.0  | 7.7  | 8.8  | 8.4  | 8.6  | 7.5  | 7.4  |  |
| DO Fathead           | mg/L     | 7.9  | 7.2  | 7.7  | 7.9  | 5.9  | 7.2  | 0.0  |  |
| pH Cerio             | S.U.     | 7.8  | 8.0  | 8.2  | 8.1  | 8.4  | 7.8  | 8.1  |  |
| pH Fathead           | S.U.     | 7.6  | 7.6  | 7.5  | 7.6  | 7.2  | 7.5  | 0.0  |  |
| Conductivity Cerio   | umhos/cm | 1100 | 1100 | 1200 | 1200 | 1200 | 1200 | 1200 |  |
| Conductivity Fathead | umhos/cm | 990  | 1100 | 1100 | 1100 | 1100 | 1100 | 0.0  |  |

# Ceriodaphnia Reproduction 208079

Title: Ceriodaphnia Reproduction City of Cadillac WWTP 208079  
 File: cr208079.txt Transform: NO TRANSFORMATION

## Kolmogorov Test for Normality

D = 0.0830 (p-value > 0.100)  
 D\* = 0.6514

Critical D\* = 1.035 (alpha = 0.01 , N = 60)  
 = 0.895 (alpha = 0.05 , N = 60)

Data PASS normality test (alpha = 0.01). Continue analysis.

## Bartlett's Test for Homogeneity of Variance

Calculated B1 statistic = 5.4058 (p-value = 0.3684)

Data PASS B1 homogeneity test at 0.01 level. Continue analysis.

Critical B = 15.0863 (alpha = 0.01, df = 5)  
 = 11.0705 (alpha = 0.05, df = 5)

## Summary Statistics on Data

TABLE 1 of 2

| GRP | IDENTIFICATION | N  | MIN     | MAX     | MEAN    |
|-----|----------------|----|---------|---------|---------|
| 1   | Control        | 10 | 28.0000 | 40.0000 | 35.4000 |
| 2   | 5%             | 10 | 29.0000 | 42.0000 | 35.1000 |
| 3   | 15%            | 10 | 31.0000 | 41.0000 | 34.8000 |
| 4   | 30%            | 10 | 27.0000 | 44.0000 | 33.1000 |
| 5   | 60%            | 10 | 31.0000 | 42.0000 | 36.7000 |
| 6   | 100%           | 10 | 27.0000 | 37.0000 | 32.9000 |

## Summary Statistics on Data

TABLE 2 of 2

| GRP | IDENTIFICATION | VARIANCE | SD     | SEM    | C.V. %  |
|-----|----------------|----------|--------|--------|---------|
| 1   | Control        | 13.6000  | 3.6878 | 1.1662 | 10.4176 |
| 2   | 5%             | 14.9889  | 3.8715 | 1.2243 | 11.0301 |
| 3   | 15%            | 7.2889   | 2.6998 | 0.8537 | 7.7580  |
| 4   | 30%            | 28.1000  | 5.3009 | 1.6763 | 16.0149 |
| 5   | 60%            | 9.5667   | 3.0930 | 0.9781 | 8.4278  |
| 6   | 100%           | 9.2111   | 3.0350 | 0.9597 | 9.2249  |

Ceriodaphnia Reproduction 208079

ANOVA Table

| SOURCE         | DF | SS       | MS      | F      |
|----------------|----|----------|---------|--------|
| Between        | 5  | 104.5333 | 20.9067 | 1.5158 |
| Within (Error) | 54 | 744.8000 | 13.7926 |        |
| Total          | 59 | 849.3333 |         |        |

(p-value = 0.2003)

Critical F = 3.3769 (alpha = 0.01, df = 5,54)  
 = 2.3861 (alpha = 0.05, df = 5,54)

Since  $F < \text{Critical } F$  FAIL TO REJECT  $H_0$ : All equal (alpha = 0.01)

Dunnett's Test - TABLE 1 OF 2 Ho:Control<Treatment

| GROUP | IDENTIFICATION | TRANSFORMED<br>MEAN | MEAN CALCULATED IN<br>ORIGINAL UNITS | T STAT  | SIG<br>0.01 |
|-------|----------------|---------------------|--------------------------------------|---------|-------------|
| 1     | Control        | 35.4000             | 35.4000                              |         |             |
| 2     | 5%             | 35.1000             | 35.1000                              | 0.1806  |             |
| 3     | 15%            | 34.8000             | 34.8000                              | 0.3613  |             |
| 4     | 30%            | 33.1000             | 33.1000                              | 1.3848  |             |
| 5     | 60%            | 36.7000             | 36.7000                              | -0.7827 |             |
| 6     | 100%           | 32.9000             | 32.9000                              | 1.5052  |             |

Dunnett critical value = 2.9900 (1 Tailed, alpha = 0.01, df [used] = 5,40)  
 (Actual df = 5,54)

Dunnett's Test - TABLE 2 OF 2 Ho:Control<Treatment

| GROUP | IDENTIFICATION | NUM OF<br>REPS | MIN SIG DIFF<br>(IN ORIG. UNITS) | % OF<br>CONTROL | DIFFERENCE<br>FROM CONTROL |
|-------|----------------|----------------|----------------------------------|-----------------|----------------------------|
| 1     | Control        | 10             |                                  |                 |                            |
| 2     | 5%             | 10             | 4.9660                           | 14.0            | 0.3000                     |
| 3     | 15%            | 10             | 4.9660                           | 14.0            | 0.6000                     |
| 4     | 30%            | 10             | 4.9660                           | 14.0            | 2.3000                     |
| 5     | 60%            | 10             | 4.9660                           | 14.0            | -1.3000                    |
| 6     | 100%           | 10             | 4.9660                           | 14.0            | 2.5000                     |



# Fathead Biomass 208079

Title: Fathead Biomass City of Cadillac WWTP 208079  
File: fh208079.txt Transform:

NO TRANSFORMATION

## Shapiro - Wilk's Test for Normality

D = 54.4328  
W = 0.6483

Critical W = 0.8840 (alpha = 0.01 , N = 24)  
W = 0.9160 (alpha = 0.05 , N = 24)

Data FAIL normality test (alpha = 0.01). Try another transformation.

Warning - The first three homogeneity tests are sensitive to non-normality and should not be performed with this data as is.

## Bartlett's Test for Homogeneity of Variance

Calculated B1 statistic = 43.2460 (p-value = 0.0000)

Data FAIL B1 homogeneity test at 0.01 level. Try another transformation.

Critical B = 15.0863 (alpha = 0.01, df = 5)  
= 11.0705 (alpha = 0.05, df = 5)

## Summary Statistics on Data

TABLE 1 of 2

| GRP | IDENTIFICATION | N | MIN     | MAX    | MEAN   |
|-----|----------------|---|---------|--------|--------|
| 1   | Control        | 4 | 3.3200  | 4.2900 | 3.6640 |
| 2   | 5%             | 4 | 4.1400  | 4.5100 | 4.3275 |
| 3   | 15%            | 4 | -4.2530 | 4.7700 | 1.4468 |
| 4   | 30%            | 4 | 3.9000  | 4.8100 | 4.2900 |
| 5   | 60%            | 4 | 3.8000  | 4.2500 | 4.0175 |
| 6   | 100%           | 4 | 3.9200  | 4.4800 | 4.1225 |

## Summary Statistics on Data

TABLE 2 of 2

| GRP | IDENTIFICATION | VARIANCE | SD     | SEM    | C.V. %   |
|-----|----------------|----------|--------|--------|----------|
| 1   | Control        | 0.1835   | 0.4284 | 0.2142 | 11.6918  |
| 2   | 5%             | 0.0324   | 0.1801 | 0.0900 | 4.1610   |
| 3   | 15%            | 17.6237  | 4.1981 | 2.0990 | 290.1716 |
| 4   | 30%            | 0.1905   | 0.4364 | 0.2182 | 10.1731  |
| 5   | 60%            | 0.0531   | 0.2304 | 0.1152 | 5.7353   |
| 6   | 100%           | 0.0611   | 0.2472 | 0.1236 | 5.9956   |

Fathead Biomass 208079

ANOVA Table

| SOURCE         | DF | SS      | MS     | F      |
|----------------|----|---------|--------|--------|
| Between        | 5  | 24.3250 | 4.8650 | 1.6088 |
| Within (Error) | 18 | 54.4328 | 3.0240 |        |
| Total          | 23 | 78.7579 |        |        |

(p-value = 0.2083)

Critical F = 4.2479 (alpha = 0.01, df = 5,18)  
 = 2.7729 (alpha = 0.05, df = 5,18)

Since  $F < \text{Critical } F$  FAIL TO REJECT  $H_0$ : All equal (alpha = 0.01)

Dunnett's Test - TABLE 1 OF 2  $H_0$ :Control<Treatment

| GROUP | IDENTIFICATION | TRANSFORMED<br>MEAN | MEAN CALCULATED IN<br>ORIGINAL UNITS | T STAT  | SIG<br>0.01 |
|-------|----------------|---------------------|--------------------------------------|---------|-------------|
| 1     | Control        | 3.6640              | 3.6640                               |         |             |
| 2     | 5%             | 4.3275              | 4.3275                               | -0.5396 |             |
| 3     | 15%            | 1.4468              | 1.4468                               | 1.8032  |             |
| 4     | 30%            | 4.2900              | 4.2900                               | -0.5091 |             |
| 5     | 60%            | 4.0175              | 4.0175                               | -0.2875 |             |
| 6     | 100%           | 4.1225              | 4.1225                               | -0.3729 |             |

Dunnett critical value = 3.2100 (1 Tailed, alpha = 0.01, df = 5,18)

Dunnett's Test - TABLE 2 OF 2  $H_0$ :Control<Treatment

| GROUP | IDENTIFICATION | NUM OF<br>REPS | MIN SIG DIFF<br>(IN ORIG. UNITS) | % OF<br>CONTROL | DIFFERENCE<br>FROM CONTROL |
|-------|----------------|----------------|----------------------------------|-----------------|----------------------------|
| 1     | Control        | 4              |                                  |                 |                            |
| 2     | 5%             | 4              | 3.9472                           | 107.7           | -0.6635                    |
| 3     | 15%            | 4              | 3.9472                           | 107.7           | 2.2173                     |
| 4     | 30%            | 4              | 3.9472                           | 107.7           | -0.6260                    |
| 5     | 60%            | 4              | 3.9472                           | 107.7           | -0.3535                    |
| 6     | 100%           | 4              | 3.9472                           | 107.7           | -0.4585                    |

Steel's Many-One Rank Test -  $H_0$ : Control<Treatment

| GROUP | IDENTIFICATION | MEAN IN<br>ORIGINAL UNITS | RANK<br>SUM | CRIT.<br>VALUE | DF   | SIG<br>0.01 |
|-------|----------------|---------------------------|-------------|----------------|------|-------------|
| 1     | Control        | 3.6640                    |             |                |      |             |
| 2     | 5%             | 4.3275                    | 24.00       | None           | 4.00 |             |
| 3     | 15%            | 1.4468                    | 18.00       | None           | 4.00 |             |
| 4     | 30%            | 4.2900                    | 24.00       | None           | 4.00 |             |
| 5     | 60%            | 4.0175                    | 22.00       | None           | 4.00 |             |
| 6     | 100%           | 4.1225                    | 23.00       | None           | 4.00 |             |

Critical values are 1 tailed ( k = 5 )

WARNING - There are no critical values for this combination  
 of groups and replicates.

Fathead Biomass 208079

PMSD >30, therefore the statistics were rerun at alpha level 0.05.

ANOVA Table

| SOURCE         | DF | SS      | MS     | F      |
|----------------|----|---------|--------|--------|
| Between        | 5  | 24.3250 | 4.8650 | 1.6088 |
| Within (Error) | 18 | 54.4328 | 3.0240 |        |
| Total          | 23 | 78.7579 |        |        |

(p-value = 0.2083)

Critical F = 4.2479 (alpha = 0.01, df = 5,18)  
 = 2.7729 (alpha = 0.05, df = 5,18)

Since F < Critical F FAIL TO REJECT Ho: All equal (alpha = 0.05)

Dunnett's Test - TABLE 1 OF 2 Ho:Control<Treatment

| GROUP | IDENTIFICATION | TRANSFORMED<br>MEAN | MEAN CALCULATED IN<br>ORIGINAL UNITS | T STAT  | SIG<br>0.05 |
|-------|----------------|---------------------|--------------------------------------|---------|-------------|
| 1     | Control        | 3.6640              | 3.6640                               |         |             |
| 2     | 5%             | 4.3275              | 4.3275                               | -0.5396 |             |
| 3     | 15%            | 1.4468              | 1.4468                               | 1.8032  |             |
| 4     | 30%            | 4.2900              | 4.2900                               | -0.5091 |             |
| 5     | 60%            | 4.0175              | 4.0175                               | -0.2875 |             |
| 6     | 100%           | 4.1225              | 4.1225                               | -0.3729 |             |

Dunnett critical value = 2.4100 (1 Tailed, alpha = 0.05, df = 5,18)

Dunnett's Test - TABLE 2 OF 2 Ho:Control<Treatment

| GROUP | IDENTIFICATION | NUM OF<br>REPS | MIN SIG DIFF<br>(IN ORIG. UNITS) | % OF<br>CONTROL | DIFFERENCE<br>FROM CONTROL |
|-------|----------------|----------------|----------------------------------|-----------------|----------------------------|
| 1     | Control        | 4              |                                  |                 |                            |
| 2     | 5%             | 4              | 2.9634                           | 80.9            | -0.6635                    |
| 3     | 15%            | 4              | 2.9634                           | 80.9            | 2.2173                     |
| 4     | 30%            | 4              | 2.9634                           | 80.9            | -0.6260                    |
| 5     | 60%            | 4              | 2.9634                           | 80.9            | -0.3535                    |
| 6     | 100%           | 4              | 2.9634                           | 80.9            | -0.4585                    |

Steel's Many-One Rank Test - Ho: Control<Treatment

| GROUP | IDENTIFICATION | MEAN IN<br>ORIGINAL UNITS | RANK<br>SUM | CRIT.<br>VALUE | DF   | SIG<br>0.05 |
|-------|----------------|---------------------------|-------------|----------------|------|-------------|
| 1     | Control        | 3.6640                    |             |                |      |             |
| 2     | 5%             | 4.3275                    | 24.00       | 10.00          | 4.00 |             |
| 3     | 15%            | 1.4468                    | 18.00       | 10.00          | 4.00 |             |
| 4     | 30%            | 4.2900                    | 24.00       | 10.00          | 4.00 |             |
| 5     | 60%            | 4.0175                    | 22.00       | 10.00          | 4.00 |             |
| 6     | 100%           | 4.1225                    | 23.00       | 10.00          | 4.00 |             |

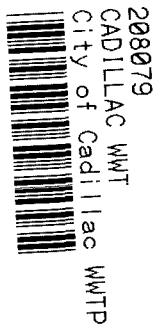
Critical values are 1 tailed ( k = 5 )



**PARAGON**  
Laboratories

**CHAIN-OF-CUSTODY**

12649 Richfield Ct. Livonia, MI  
P 734.462.3900 F 734.462.3911 W www.p



ge \_\_\_\_ of \_\_\_\_

|                                                                                                                                                                                              |                  |                   |               |              |                                                              |                                                                        |                                                                   |                    |          |          |          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------|---------------|--------------|--------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------|----------|----------|----------|
| Client Name: <u>Cadillac WWT</u>                                                                                                                                                             |                  |                   |               |              |                                                              | INTERNAL USE ONLY                                                      |                                                                   |                    |          |          |          |
| Contact Person: <u>Donny W.</u>                                                                                                                                                              |                  |                   |               |              |                                                              | Paragon Report No.: <u>208079</u>                                      |                                                                   |                    |          |          |          |
| Mailing Address: <u>3000 N. Lake St</u>                                                                                                                                                      |                  |                   |               |              |                                                              | Logged By: <u>RES</u>                                                  |                                                                   |                    |          |          |          |
| City, State, Zip: <u>Cadillac, MI 49601</u>                                                                                                                                                  |                  |                   |               |              |                                                              | Sample Condition Upon Receipt: <u>Acceptable</u> Other (Specify Below) |                                                                   |                    |          |          |          |
| Phone and Fax: <u>(231) 725-2368 / (231) 725-0875</u>                                                                                                                                        |                  |                   |               |              |                                                              | IR Temperature Upon Receipt: <u>4°C</u>                                |                                                                   |                    |          |          |          |
| Email: <u>lch@cadillac-mi.net</u>                                                                                                                                                            |                  |                   |               |              |                                                              | Remarks: <u>sample received chilled on natural ice - RES S-5-16</u>    |                                                                   |                    |          |          |          |
| Client Job Name / No.:                                                                                                                                                                       |                  |                   |               |              |                                                              |                                                                        |                                                                   |                    |          |          |          |
| Job Location:                                                                                                                                                                                |                  |                   |               |              |                                                              |                                                                        |                                                                   |                    |          |          |          |
| Sampled By: <u>DS</u>                                                                                                                                                                        |                  |                   |               |              |                                                              | PO No.:                                                                |                                                                   |                    |          |          |          |
| WHERE WAS THE SAMPLE(S) TAKEN:<br><div>Facility Does Not Chlorinate    After Chlorination    After Chlorination, Before Dechlorination<br/>After Dechlorination    Before Chlorination</div> |                  |                   |               |              |                                                              |                                                                        |                                                                   |                    |          |          |          |
| WAS THE SAMPLE(S) DECHLORINATED: YES NO IF YES, PLEASE DESCRIBE DECHLORINATION:                                                                                                              |                  |                   |               |              |                                                              |                                                                        |                                                                   |                    |          |          |          |
| WAS THE SAMPLE(S) FILTERED: YES <u>NO</u> IF YES, PLEASE STATE THE FILTER MESH SIZE:                                                                                                         |                  |                   |               |              |                                                              |                                                                        |                                                                   |                    |          |          |          |
| Item No.                                                                                                                                                                                     | Client Sample ID | No. of containers | Grab          | No. of Grabs | Grab Sample Period (Start Date and Time - End Date and Time) | Comp                                                                   | Composite Sample Period (Start Date and Time - End Date and Time) | ANALYSIS REQUESTED |          |          |          |
| 1                                                                                                                                                                                            | <u>ETH/6041</u>  | <u>2</u>          |               |              |                                                              | <u>X</u>                                                               | <u>5/13/16 7:00am - 5/14/16 12:00pm</u>                           | <u>X</u>           | <u>X</u> | <u>X</u> | <u>X</u> |
| 2                                                                                                                                                                                            |                  |                   |               |              |                                                              |                                                                        |                                                                   |                    |          |          |          |
|                                                                                                                                                                                              |                  |                   |               |              |                                                              |                                                                        |                                                                   |                    |          |          |          |
|                                                                                                                                                                                              |                  |                   |               |              |                                                              |                                                                        |                                                                   |                    |          |          |          |
|                                                                                                                                                                                              |                  |                   |               |              |                                                              |                                                                        |                                                                   |                    |          |          |          |
|                                                                                                                                                                                              |                  |                   |               |              |                                                              |                                                                        |                                                                   |                    |          |          |          |
| Tran. #                                                                                                                                                                                      | Released By      | Received By       | Date          | Time         | Tran. #                                                      | Released By                                                            | Received By                                                       | Date               | Time     |          |          |
| 1.                                                                                                                                                                                           | <u>NAV</u>       | <u>UPS</u>        | <u>5-5-16</u> | <u>9:32</u>  | 3.                                                           |                                                                        |                                                                   |                    |          |          |          |
| 2.                                                                                                                                                                                           |                  |                   |               |              | 4.                                                           |                                                                        |                                                                   |                    |          |          |          |



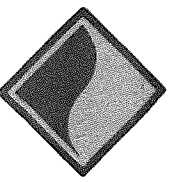
**PARAGON**  
Laboratories

# CHAIN-OF-CUSTODY RECORD

12649 Richfield Ct. Livonia, MI 48150  
P 734.462.3900 F 734.462.3911 W www.paragonlaboratories.com

Page \_\_\_\_ of \_\_\_\_

|                                                                                 |                  |                   |               |                                                                        |                                                              |                                           |                                                                   |                                                        |                                                        |
|---------------------------------------------------------------------------------|------------------|-------------------|---------------|------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|
| Client Name: <u>Cadillac Water</u>                                              |                  |                   |               | INTERNAL USE ONLY                                                      |                                                              |                                           |                                                                   |                                                        |                                                        |
| Contact Person: <u>Angela</u>                                                   |                  |                   |               | Paragon Report No.: <u>208074</u>                                      |                                                              |                                           |                                                                   |                                                        |                                                        |
| Mailing Address: <u>300 N. Lake St</u>                                          |                  |                   |               | Logged By: <u>PES</u>                                                  |                                                              |                                           |                                                                   |                                                        |                                                        |
| City, State, Zip: <u>Cadillac, MI 49601</u>                                     |                  |                   |               | Sample Condition Upon Receipt: <u>acceptable</u> Other (Specify Below) |                                                              |                                           |                                                                   |                                                        |                                                        |
| Phone and Fax: <u>(231) 775-2368 / (231) 775-2876</u>                           |                  |                   |               | IR Temperature Upon Receipt: <u>40C</u>                                |                                                              |                                           |                                                                   |                                                        |                                                        |
| Email: <u>lab@cadillac-water.com</u>                                            |                  |                   |               | Remarks: <u>sample received chilled on natural ice - PES 5-6-16</u>    |                                                              |                                           |                                                                   |                                                        |                                                        |
| Client Job Name / No.:                                                          |                  |                   |               |                                                                        |                                                              |                                           |                                                                   |                                                        |                                                        |
| Job Location:                                                                   |                  | PO No.:           |               |                                                                        |                                                              |                                           |                                                                   |                                                        |                                                        |
| Sampled By: <u>JS</u>                                                           |                  |                   |               |                                                                        |                                                              |                                           |                                                                   |                                                        |                                                        |
| WHERE WAS THE SAMPLE(S) TAKEN: <u>Facility Does Not Chlorinate</u>              |                  |                   |               | After Chlorination                                                     |                                                              | After Chlorination, Before Dechlorination |                                                                   |                                                        |                                                        |
|                                                                                 |                  |                   |               | After Dechlorination                                                   |                                                              | Before Chlorination                       |                                                                   |                                                        |                                                        |
| WAS THE SAMPLE(S) DECHLORINATED: YES NO IF YES, PLEASE DESCRIBE DECHLORINATION: |                  |                   |               |                                                                        |                                                              |                                           |                                                                   |                                                        |                                                        |
| WAS THE SAMPLE(S) FILTERED: YES NO IF YES, PLEASE STATE THE FILTER MESH SIZE:   |                  |                   |               |                                                                        |                                                              |                                           |                                                                   |                                                        |                                                        |
| Item No.                                                                        | Client Sample ID | No. of containers | Grab          | No. of Grabs                                                           | Grab Sample Period (Start Date and Time - End Date and Time) | Comp                                      | Composite Sample Period (Start Date and Time - End Date and Time) | ANALYSIS REQUESTED                                     |                                                        |
|                                                                                 | <u>EHL-1001</u>  | <u>2</u>          |               |                                                                        |                                                              | <input checked="" type="checkbox"/>       | <u>SYNTH. TREAT. - 5/5/16 TUES</u>                                | <input checked="" type="checkbox"/> Ceriodaphnia Dubia | <input checked="" type="checkbox"/> Fathead Minnows    |
|                                                                                 |                  |                   |               |                                                                        |                                                              |                                           |                                                                   | <input checked="" type="checkbox"/> Daphnia Magna      | <input checked="" type="checkbox"/> Ceriodaphnia Dubia |
|                                                                                 |                  |                   |               |                                                                        |                                                              |                                           |                                                                   | <input checked="" type="checkbox"/> Fathead Minnows    | <input checked="" type="checkbox"/>                    |
|                                                                                 |                  |                   |               |                                                                        |                                                              |                                           |                                                                   |                                                        |                                                        |
|                                                                                 |                  |                   |               |                                                                        |                                                              |                                           |                                                                   |                                                        |                                                        |
|                                                                                 |                  |                   |               |                                                                        |                                                              |                                           |                                                                   |                                                        |                                                        |
|                                                                                 |                  |                   |               |                                                                        |                                                              |                                           |                                                                   |                                                        |                                                        |
|                                                                                 |                  |                   |               |                                                                        |                                                              |                                           |                                                                   |                                                        |                                                        |
| Tran. #                                                                         | Released By      | Received By       | Date          | Time                                                                   | Tran. #                                                      | Released By                               | Received By                                                       | Date                                                   | Time                                                   |
| 1.                                                                              | <u>JS</u>        | <u>JS</u>         | <u>5-6-16</u> | <u>9:30</u>                                                            | 3.                                                           |                                           |                                                                   |                                                        |                                                        |
| 2.                                                                              | <u>JS</u>        | <u>JS</u>         | <u>5-6-16</u> | <u>9:30</u>                                                            | 4.                                                           |                                           |                                                                   |                                                        |                                                        |



**PARAGON**  
Laboratories

# CHAIN-OF-CUSTODY RECORD

12649 Richfield Ct. Livonia, MI 48150  
P 734.462.3900 F 734.462.3911 W www.paragonlaboratories.com

Page \_\_\_\_ of \_\_\_\_

|                                                                                                                                                      |                  |                   |                |                                                                        |                                                              |             |                                                                   |                    |                 |               |                    |                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------|----------------|------------------------------------------------------------------------|--------------------------------------------------------------|-------------|-------------------------------------------------------------------|--------------------|-----------------|---------------|--------------------|-----------------|
| Client Name: <u>Cadillac Water</u>                                                                                                                   |                  |                   |                | INTERNAL USE ONLY                                                      |                                                              |             |                                                                   |                    |                 |               |                    |                 |
| Contact Person: <u>Barry V. I.</u>                                                                                                                   |                  |                   |                | Paragon Report No.: <u>208079</u>                                      |                                                              |             |                                                                   |                    |                 |               |                    |                 |
| Mailing Address: <u>300 N. Lake St</u>                                                                                                               |                  |                   |                | Logged By: <u>RES</u>                                                  |                                                              |             |                                                                   |                    |                 |               |                    |                 |
| City, State, Zip: <u>Cadillac, MI 49601</u>                                                                                                          |                  |                   |                | Sample Condition Upon Receipt: <u>Acceptable</u> Other (Specify Below) |                                                              |             |                                                                   |                    |                 |               |                    |                 |
| Phone and Fax: <u>(231) 225-3688 / (231) 225-2872</u>                                                                                                |                  |                   |                | IR Temperature Upon Receipt: <u>3°C</u>                                |                                                              |             |                                                                   |                    |                 |               |                    |                 |
| Email: <u>label@cadillac-water.com</u>                                                                                                               |                  |                   |                | Remarks: <u>sample received chilled on natural ice - RES 5-10-16</u>   |                                                              |             |                                                                   |                    |                 |               |                    |                 |
| Client Job Name / No.:                                                                                                                               |                  |                   |                |                                                                        |                                                              |             |                                                                   |                    |                 |               |                    |                 |
| Job Location:                                                                                                                                        |                  |                   |                |                                                                        |                                                              |             |                                                                   |                    |                 |               |                    |                 |
| Sampled By: <u>JS</u> PO No.:                                                                                                                        |                  |                   |                |                                                                        |                                                              |             |                                                                   |                    |                 |               |                    |                 |
| WHERE WAS THE SAMPLE(S) TAKEN: <u>Facility Does Not Chlorinate</u> After Chlorination After Dechlorination Before Chlorination Before Dechlorination |                  |                   |                | ANALYSIS REQUESTED                                                     |                                                              |             |                                                                   |                    |                 |               |                    |                 |
| WAS THE SAMPLE(S) DECHLORINATED: YES NO IF YES, PLEASE DESCRIBE DECHLORINATION:                                                                      |                  |                   |                | Acute Toxicity Chronic Toxicity                                        |                                                              |             |                                                                   |                    |                 |               |                    |                 |
| WAS THE SAMPLE(S) FILTERED: YES <u>NO</u> IF YES, PLEASE STATE THE FILTER MESH SIZE:                                                                 |                  |                   |                |                                                                        |                                                              |             |                                                                   |                    |                 |               |                    |                 |
| Item No.                                                                                                                                             | Client Sample ID | No. of containers | Grab           | No. of Grabs                                                           | Grab Sample Period (Start Date and Time - End Date and Time) | Comp        | Composite Sample Period (Start Date and Time - End Date and Time) | Ceriodaphnia Dubia | Fathead Minnows | Daphnia Magna | Ceriodaphnia Dubia | Fathead Minnows |
| 1                                                                                                                                                    | <u>EFFluent</u>  | <u>2</u>          |                |                                                                        |                                                              | <u>X</u>    | <u>5/10/16 5:15 AM - 7:15 PM</u>                                  | <u>X</u>           | <u>X</u>        |               | <u>X</u>           | <u>X</u>        |
|                                                                                                                                                      |                  |                   |                |                                                                        |                                                              |             |                                                                   |                    |                 |               |                    |                 |
|                                                                                                                                                      |                  |                   |                |                                                                        |                                                              |             |                                                                   |                    |                 |               |                    |                 |
|                                                                                                                                                      |                  |                   |                |                                                                        |                                                              |             |                                                                   |                    |                 |               |                    |                 |
|                                                                                                                                                      |                  |                   |                |                                                                        |                                                              |             |                                                                   |                    |                 |               |                    |                 |
|                                                                                                                                                      |                  |                   |                |                                                                        |                                                              |             |                                                                   |                    |                 |               |                    |                 |
| Tran. #                                                                                                                                              | Released By      | Received By       | Date           | Time                                                                   | Tran. #                                                      | Released By | Received By                                                       | Date               | Time            |               |                    |                 |
| 1.                                                                                                                                                   | <u>RES</u>       |                   |                |                                                                        | 3.                                                           |             |                                                                   |                    |                 |               |                    |                 |
| 2.                                                                                                                                                   | <u>RES</u>       | <u>SCJ</u>        | <u>5-10-16</u> | <u>9:50</u>                                                            | 4.                                                           |             |                                                                   |                    |                 |               |                    |                 |